1. Finalni predpoved ceny:

```
\widehat P_{i,t}^{fut}(T_\ell)=\sum_{s=1}^{N_s}p_s\,\widetilde P_{i,t}^{(s)}(T_\ell)
```

Copy

$$\widehat{P}_{i,t}^{\,fut}(T_\ell) = \sum_{s=1}^{N_s} p_s \, \widetilde{P}_{i,t}^{(s)}(T_\ell)$$

s = jeden ze scenaru nebo traektorii

p_s = pravdepodobnost scenare

\widetilde P_{i,t} = predpokladana cena ve scenari “s”

\sum_s p_s = 1

2. Cena uvnitr scenare:

```
\widetilde P_{i,t}^{(s)}(E_{i,\ell})=\exp\left[\widetilde p_{i,t}^{(s)}(E_{i,\ell})\right],
\\
\begin{aligned}\widetilde p_{i,t}^{(s)}=
{}&f_{i,t}^{mkt,P}+\alpha_{i,\ell}\left(\ln P_{i,t}^{str,(0)}-f_{i,t}^{mkt,P}\right)+\left(\ln P_{i,t}^{str,(s)}-\ln P_{i,t}^{str,(0)}\right)\end{aligned}
```

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$$\widetilde{P}_{i,t}^{(s)}(E_{i,\ell}) = \exp\left[\tilde{p}_{i,t}^{(s)}(E_{i,\ell})\right],$$

$$\tilde{p}_{i,t}^{(s)} = f_{i,t}^{mkt,P} + \alpha_{i,\ell}\left(\ln P_{i,t}^{str,(0)} - f_{i,t}^{mkt,P}\right) + \left(\ln P_{i,t}^{str,(s)} - \ln P_{i,t}^{str,(0)}\right)$$

\alpha = vaha “strukturniho modelu”

Stukturni model (\alpha):

```
\alpha_{i,\ell}=\frac{(\sigma_{i,\ell}^{mkt})^2}
{(\sigma_{i,\ell}^{mkt})^2+(\sigma_{i,\ell}^{str})^2}
```

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$$\alpha_{i,\ell} = \frac{(\sigma_{i,\ell}^{mkt})^2}{(\sigma_{i,\ell}^{mkt})^2 + (\sigma_{i,\ell}^{str})^2}$$

\sigma^{mkt} = nejistota trzniho oceneni

\sigma^{str} = nejistota stukturniho modelu

Strukturni predpoved (P^{str}):

```
P_{i,t}^{str,(s)}(E)=\mathbb E_t^{P,
(s)}\left[\exp\left(m_{i,E}^{(s)}+x_{i,E}^{(s)}\right)\right]
```

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$$P_{i,t}^{str,(s)}(E) = \mathbb{E}_t^{P,(s)}\left[\exp\left(m_{i,E}^{(s)} + x_{i,E}^{(s)}\right)\right]$$

Pri Monte-Karlo:

```
P_{i,t}^{str,(s)}
(E)\approx\frac1N\sum_{n=1}^{N}\exp(m_{i,E}^{(s,n)}+x_{i,E}^
{(s,n)})
```

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$$P_{i,t}^{str,(s)}(E) \approx \frac{1}{N}\sum_{n=1}^{N}\exp(m_{i,E}^{(s,n)} + x_{i,E}^{(s,n)})$$

m = zakladni vrstva

x = trzni vrstva

Spot cena:

```
\mathbf P_t=\exp(\mathbf m_t+\mathbf x_t)
```

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$$\mathbf{P}_t = \exp(\mathbf{m}_t + \mathbf{x}_t)$$

Zakladni vrstva:

```
d\mathbf m_t=\mathbf K_m\left[\mathbf M(\mathbf A_t,\mathbf
D_t,R_t)-\mathbf m_t\right]dt+\mathbf B_L\,d\mathbf
Q^{(L)}_t+\mathbf\Sigma_m\,d\mathbf W_t^m
```

Copy

$$d\mathbf{m}_t = \mathbf{K}_m \left[\mathbf{M}(\mathbf{A}_t, \mathbf{D}_t, R_t) - \mathbf{m}_t\right] dt + \mathbf{B}_L \, d\mathbf{Q}_t^{(L)} + \mathbf{\Sigma}_m \, d\mathbf{W}_t^m$$

Trzni vrstva:

```
d\mathbf x_t=[-\mathbf K_{z_t}\mathbf x_t+\mathbf
B_{z_t}\mathbf u_t+\mathbf G_{z_t}\mathbf x_t]dt+\mathbf
B_Td\mathbf Q_t^{(T)}+\mathbf\Sigma_{z_t}d\mathbf W_t^x
```

Copy

$$d\mathbf{x}_t = [-\mathbf{K}_{z_t}\mathbf{x}_t + \mathbf{B}_{z_t}\mathbf{u}_t + \mathbf{G}_{z_t}\mathbf{x}_t]dt + \mathbf{B}_T d\mathbf{Q}_t^{(T)} + \mathbf{\Sigma}_{z_t} d\mathbf{W}_t^x$$

u_t:

```
\mathbf u_t^{(h)}=\begin{bmatrix}2\mathbf A_t^{(h)}-
\mathbf1\\2\mathbf D_t-\mathbf1\\(2R_t-
1)\mathbf1\\\boldsymbol{\mathcal S}_t^{(h)}\end{bmatrix}
```

Copy

$$\mathbf{u}_t^{(h)} = \begin{bmatrix} 2\mathbf{A}_t^{(h)} - \mathbf{1} \\ 2\mathbf{D}_t - \mathbf{1} \\ (2R_t - 1)\mathbf{1} \\ \boldsymbol{\mathcal{S}}_t^{(h)} \end{bmatrix}$$

Zakladni “target” cena:

```
M_{i,t}=\sum_{k=1}^{K}\pi_{k,t}M^{(k)}_{i,t}
```

Copy

$$M_{i,t} = \sum_{k=1}^{K} \pi_{k,t} M_{i,t}^{(k)}$$

Vnitrni normalizace:

```
a_{i,t}=2A_{i,t}-1, \qquad d_{i,t}=2D_{i,t}-1,\qquad r_t=2R_t-1
```

Copy

$$a_{i,t} = 2A_{i,t} - 1, \qquad d_{i,t} = 2D_{i,t} - 1, \qquad r_t = 2R_t - 1$$

Rezim “k”:

```
\begin{aligned}M^{(k)}_{i,t}=
{}&\mu_{i,k}+\delta_{i,k}t+\theta^{(k)}_{A,i}a_{i,t}+\theta^{(k)}_{D,i}d_{i,t}+\theta^{(k)}_{R,i}r_t
\\
&+\theta^{(k)}_{AD,i}a_{i,t}d_{i,t}+\theta^{(k)}_{AR,i}a_{i,t}r_t+\theta^{(k)}_{DR,i}d_{i,t}r_t\end{aligned}
```

Copy

$$\begin{aligned}M^{(k)}_{i,t} = {}&\mu_{i,k} + \delta_{i,k}t + \theta^{(k)}_{A,i}a_{i,t} + \theta^{(k)}_{D,i}d_{i,t} + \theta^{(k)}_{R,i}r_t \\ &+ \theta^{(k)}_{AD,i}a_{i,t}d_{i,t} + \theta^{(k)}_{AR,i}a_{i,t}r_t + \theta^{(k)}_{DR,i}d_{i,t}r_t\end{aligned}$$

Trzni krivka:

```
\begin{aligned}f_{i,t}^{mkt}(\tau)=\underset{f:f(0)=\ln P_{i,t}}
{\arg\min}\Bigg\
{&\sum_{j=1}^{2}w_{i,j}^{liq}\left[f(\tau_{i,j})-\ln
P_{i,t}^{fut,(j)}\right]^2
\\
&+\lambda_{carry}\int_0^{\tau_{max}}\left[f'(v)-
c^{carry}_{i,t+v}\right]^2dv
\\
&+\lambda_{smooth}\int_0^{\tau_{max}}\left[f''(v)\right]^2dv\
```

Copy

$$f_{i,t}^{mkt}(\tau) = \underset{f:f(0)=\ln P_{i,t}}{\arg\min} \left\{ \sum_{j=1}^{2} w_{i,j}^{liq} \left[f(\tau_{i,j}) - \ln P_{i,t}^{fut,(j)} \right]^2 + \lambda_{carry} \int_0^{\tau_{max}} \left[f'(v) - c_{i,t+v}^{carry} \right]^2 dv + \lambda_{smooth} \int_0^{\tau_{max}} [f''(v)]^2 dv \right\}$$

P_{i,t} = spot cena

P^{fut,(j) (de facto (1) a (2)) = dva aktualni futures

\tau_{i,j} = doba do expirace kazdeho futures

Ocekavana budouci cena bez premie za risk:

```
f_{i,t}^{mkt,P}(\tau)=f_{i,t}^{mkt}(\tau)-RP_{i,t}(\tau)
```

Copy

$$f_{i,t}^{mkt,P}(\tau) = f_{i,t}^{mkt}(\tau) - RP_{i,t}(\tau)$$

Premie za risk:

```
RP_{i,t}(\tau)=f_{i,t}^{mkt}(\tau)-\ln\mathbb
E_t^P\left[P_{i,t+\tau}\right]
```

Copy

$$RP_{i,t}(\tau) = f_{i,t}^{mkt}(\tau) - \ln \mathbb{E}_t^P \left[P_{i,t+\tau}\right]$$

Vyhoda mit presnou komoditu fyzicky:

```
y_{i,t}=\operatorname{clip}_{[y_i^{min},y_i^{max}]}\left[y_{i,0
}+\gamma_{D,i}D_{i,t}+\gamma_{I,i}q_{amount,i,t}-
\gamma_{B,i}B_{i,t}+\sum_{k=1}^{K}\pi_{k,t}\gamma_{i,k}^{re
g}\right]
```

Copy

$$y_{i,t} = \operatorname{clip}_{[y_i^{min}, y_i^{max}]} \left[y_{i,0} + \gamma_{D,i} D_{i,t} + \gamma_{I,i} q_{amount,i,t} - \gamma_{B,i} B_{i,t} + \sum_{k=1}^{K} \pi_{k,t} \gamma_{i,k}^{reg} \right]$$

Cena presunu komodity:

```
c^{carry}_{i,t}=r_t+c^{storage}_{i,t}+c^{insurance}_{i,t}-
y_{i,t}
```

Copy

$$c_{i,t}^{carry} = r_t + c_{i,t}^{storage} + c_{i,t}^{insurance} - y_{i,t}$$